HORIZON MEDICAL — AI/ML Documentation

Documento: HM-AIML-001
Versión: 1.0
Fecha: 2026-05-16
Referencia: FDA AI/ML-Based SaMD Guidance, GMLP, IMDRF SaMD Framework

Tabla de Contenidos

1. Regulatory Framework for AI/ML

1.1 FDA AI/ML Framework

Document	Relevance
AI/ML-Based SaMD Action Plan (2021)	Overall FDA strategy for regulating AI/ML in medical devices
Predetermined Change Control Plan (PCCP) Guidance (2023)	Framework for planned modifications to AI/ML algorithms
Clinical Decision Support Guidance	Criteria for CDS exclusion from device regulation
Software as Medical Device (SaMD) Guidance	IMDRF framework for SaMD categorization
Good Machine Learning Practice (GMLP) Guiding Principles	10 principles for developing safe and effective AI/ML
Marketing Submission for AI/ML-enabled Device Software Functions (2025)	Guidance on documentation requirements for 510(k) submissions

1.2 SaMD Categorization (IMDRF)

Factor	Horizon Medical AI Module
Significance of information	Treat or diagnose → "Inform clinical management" (the AI provides arrhythmia classification that informs the clinician's decision)
Healthcare situation or condition	Serious (cardiac arrhythmias — VT/VF can be life-threatening)
SaMD Category	**Category III** (Serious condition + Inform clinical management)

1.3 Locked vs. Adaptive Algorithm

Type	Horizon Medical
Algorithm type	**Locked algorithm**
Definition	The algorithm does not learn or modify its behavior after deployment. Any changes to the model require a new version release through the change control process.
Justification	A locked algorithm provides the most predictable and auditable behavior for regulatory compliance. Future versions with adaptive capabilities would require a PCCP.

---

2. Algorithm Description

2.1 Model Architecture: CNN-LSTM Hybrid

```
Input: ECG Segment (10 seconds × 3 channels × 500 Hz = 15,000 samples)
    │
    ▼
┌─────────────────────────────────────────┐
│          PREPROCESSING                  │
│  • Resampling to 250 Hz                 │
│  • Bandpass filter (0.5–45 Hz)          │
│  • Z-score normalization                │
│  • Segmentation (10s windows)           │
└─────────────────────────────────────────┘
               │
               ▼
┌─────────────────────────────────────────┐
│     CNN FEATURE EXTRACTOR               │
│                                         │
│  Conv1D (64 filters, k=7) + BN + ReLU   │
│  MaxPool1D (2)                          │
│  Conv1D (128 filters, k=5) + BN + ReLU  │
│  MaxPool1D (2)                          │
│  Conv1D (256 filters, k=3) + BN + ReLU  │
│  MaxPool1D (2)                          │
│  Conv1D (256 filters, k=3) + BN + ReLU  │
│  Global Average Pooling                 │
│                                         │
│  Output: Feature vector (256-dim)       │
└─────────────────────────────────────────┘
               │
               ▼
┌─────────────────────────────────────────┐
│     LSTM TEMPORAL ANALYZER              │
│                                         │
│  LSTM (128 units, bidirectional)        │
│  Dropout (0.3)                          │
│  LSTM (64 units, bidirectional)         │
│  Dropout (0.3)                          │
│                                         │
│  Output: Temporal features (128-dim)    │
└─────────────────────────────────────────┘
               │
               ▼
┌─────────────────────────────────────────┐
│     CLASSIFIER                          │
│                                         │
│  Dense (64) + ReLU + Dropout (0.3)      │
│  Dense (N_classes) + Softmax            │
│                                         │
│  Output: Class probabilities            │
└─────────────────────────────────────────┘
               │
               ▼
┌─────────────────────────────────────────┐
│     POST-PROCESSING                     │
│                                         │
│  • Confidence thresholding              │
│  • Temporal aggregation (majority       │
│    voting over consecutive segments)    │
│  • Alert priority assignment            │
│                                         │
│  Output: Arrhythmia label + priority    │
└─────────────────────────────────────────┘
```

2.2 Output Classes

Class ID	Arrhythmia	Alert Priority
0	Normal Sinus Rhythm (NSR)	None
1	Atrial Fibrillation (AF)	High
2	Atrial Flutter (AFL)	High
3	Ventricular Tachycardia (VT)	Critical
4	Supraventricular Tachycardia (SVT)	Medium
5	Sinus Bradycardia	Medium
6	Sinus Tachycardia	Low
7	Premature Ventricular Complex (PVC)	Low–Medium
8	Premature Atrial Complex (PAC)	Low
9	AV Block (2nd/3rd degree)	High
10	Sinus Pause (>3 seconds)	High

2.3 Model Hyperparameters

Parameter	Value
Input size	10s × 3ch × 250 Hz = 7,500 samples
CNN layers	4 Conv1D layers (64, 128, 256, 256 filters)
LSTM layers	2 bidirectional LSTM layers (128, 64 units)
Dropout rate	0.3
Optimizer	Adam (lr=1e-4, weight decay=1e-5)
Loss function	Weighted cross-entropy (class weights inversely proportional to frequency)
Batch size	64
Training epochs	100 (with early stopping, patience=10)
Total parameters	~2.5M
Inference time	<1 second per 10-second segment (GPU); <3 seconds (CPU)

3. Training Data and Validation

3.1 Training Data Sources

Dataset	Source	Size	Annotations	Use
MIT-BIH Arrhythmia Database	PhysioNet	48 records, 30 min each	Beat-level by cardiologists	Training + validation
MIT-BIH AF Database	PhysioNet	25 records, 10h each	AF episodes annotated	Training + validation
INCART Database	PhysioNet	75 records, 30 min each	Beat-level annotations	Training + validation
PTB-XL	PhysioNet	21,837 records, 10s each	Multi-label by cardiologists	Training + validation
Chapman-Shaoxing	PhysioNet	10,646 records	Multi-label	Training + validation
Proprietary clinical dataset	[Hospital partner]	[N] records, 24h each	Board-certified cardiologists (2 reviewers + adjudicator)	Validation (held-out)

3.2 Data Split

Split	Proportion	Purpose	Records
Training	70%	Model training	[N]
Validation	15%	Hyperparameter tuning, early stopping	[N]
Test (internal)	15%	Performance evaluation	[N]
Test (external)	Separate	Independent validation	[N] (proprietary clinical dataset)

> **Important:** The external test set was NOT used during any stage of model development (training, tuning, architecture selection). It was reserved exclusively for final performance evaluation.

3.3 Data Quality Controls

Control	Description
Annotation protocol	Double annotation by board-certified cardiologists; disagreements resolved by third adjudicator
Inter-annotator agreement	Cohen's kappa ≥ 0.85 required
Signal quality check	Automated quality assessment; segments with >50% artifact excluded
Data augmentation	Time warping, amplitude scaling, noise injection, baseline wander — applied only to training set
Class balancing	Weighted sampling + SMOTE for underrepresented classes

3.4 Demographic Distribution of Training Data

Demographic Factor	Distribution	Notes
Age	18–30: [%], 31–50: [%], 51–70: [%], 71+: [%]	Weighted toward 51–70 (higher arrhythmia prevalence)
Sex	Male: [%], Female: [%]	Approximately balanced
Race/Ethnicity	Caucasian: [%], African American: [%], Asian: [%], Hispanic: [%], Other: [%]	Dataset-dependent; bias analysis performed
Comorbidities	Hypertension: [%], Diabetes: [%], Heart failure: [%], CAD: [%]	Represented in clinical dataset

4. Performance Metrics

4.1 Overall Performance (Internal Test Set)

Arrhythmia	N (test)	Sensitivity	Specificity	PPV	NPV	AUC-ROC	F1
AF	[N]	[%]	[%]	[%]	[%]	[value]	[value]
AFL	[N]	[%]	[%]	[%]	[%]	[value]	[value]
VT	[N]	[%]	[%]	[%]	[%]	[value]	[value]
SVT	[N]	[%]	[%]	[%]	[%]	[value]	[value]
Bradycardia	[N]	[%]	[%]	[%]	[%]	[value]	[value]
Tachycardia	[N]	[%]	[%]	[%]	[%]	[value]	[value]
PVC	[N]	[%]	[%]	[%]	[%]	[value]	[value]
PAC	[N]	[%]	[%]	[%]	[%]	[value]	[value]
AV Block	[N]	[%]	[%]	[%]	[%]	[value]	[value]
Sinus Pause	[N]	[%]	[%]	[%]	[%]	[value]	[value]

4.2 Performance on External Validation Set

Arrhythmia	N	Sensitivity	Specificity	AUC-ROC
AF	[N]	[%]	[%]	[value]
VT	[N]	[%]	[%]	[value]
[Other arrhythmias]	[N]	[%]	[%]	[value]

4.3 Confusion Matrix (Example — Internal Test Set)

```
              Predicted →
              NSR   AF   AFL   VT   SVT  Brady Tachy PVC  PAC  AVB  Pause
Actual ↓
NSR          [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
AF           [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
AFL          [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
VT           [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
SVT          [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
Bradycardia  [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
Tachycardia  [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
PVC          [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
PAC          [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
AV Block     [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
Sinus Pause  [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ]  [  ] [  ] [  ] [  ]
```

4.4 Robustness Testing

Condition	Performance Impact	Acceptable?
Noisy signal (SNR = 10 dB)	Sensitivity drops by [%]	☐ Yes / ☐ No
Motion artifact (simulated walking)	Sensitivity drops by [%]	☐ Yes / ☐ No
Electrode partial detachment	Sensitivity drops by [%]	☐ Yes / ☐ No
Baseline wander	Sensitivity drops by [%]	☐ Yes / ☐ No
Low amplitude ECG (0.5× normal)	Sensitivity drops by [%]	☐ Yes / ☐ No

5. Bias and Fairness Analysis

5.1 Subgroup Performance Analysis

Subgroup	N (test)	AF Sensitivity	VT Sensitivity	Overall AUC
Age 18–40	[N]	[%]	[%]	[value]
Age 41–65	[N]	[%]	[%]	[value]
Age 65+	[N]	[%]	[%]	[value]
Male	[N]	[%]	[%]	[value]
Female	[N]	[%]	[%]	[value]
Caucasian	[N]	[%]	[%]	[value]
African American	[N]	[%]	[%]	[value]
Asian	[N]	[%]	[%]	[value]
Hispanic	[N]	[%]	[%]	[value]
BMI < 25	[N]	[%]	[%]	[value]
BMI ≥ 30	[N]	[%]	[%]	[value]

5.2 Fairness Metrics

Metric	Target	Result
Equalized odds (sensitivity difference across demographics)	< 5% absolute difference	[Result]
Demographic parity (positive prediction rate across groups)	< 5% absolute difference	[Result]
Calibration (reliability across subgroups)	Calibration slope 0.8–1.2 in all subgroups	[Result]

5.3 Bias Mitigation Measures

1. **Training data diversity:** Ensured representation of major demographic groups
2. **Class-weighted loss:** Prevents bias toward majority class
3. **Subgroup validation:** Separate performance evaluation per demographic
4. **Data augmentation:** Applied uniformly across groups

5. **Ongoing monitoring:** Post-market monitoring of performance by demographics (PMCF Plan)

5.4 Known Limitations

Limitation	Description	Mitigation
Underrepresentation of pediatric data	Model not validated for patients <18 years	Contraindication in labeling
Limited data from certain ethnicities	Some ethnic groups underrepresented in training data	Ongoing data collection in PMCF
Pacemaker/ICD patients	Model not trained on paced rhythms	Warning in labeling

---

6. Continuous Learning Plan

6.1 Current Status: Locked Algorithm

The Horizon Medical AI Module v1.0 is deployed as a **locked algorithm**. The model weights, architecture, and inference logic are fixed at the time of deployment and do not change based on new data or user interactions.

6.2 Update Process

Step	Description	Regulatory Impact
1	Data collection from post-market monitoring	No regulatory impact
2	Model retraining (offline, on new dataset)	Internal activity
3	Performance comparison (new vs. current)	Internal activity
4	Regulatory assessment of changes	Determine if new 510(k) or PCCP applies
5	V&V of new model version	Required
6	Regulatory submission (if required)	510(k) or PCCP update
7	Deployment of new version via OTA (backend only)	Controlled release

6.3 Future Consideration: PCCP

If Horizon Medical decides to implement adaptive AI in future versions, a **Predetermined Change Control Plan (PCCP)** will be developed following FDA Guidance, specifying:
- Types of changes anticipated (e.g., threshold adjustments, additional arrhythmia classes)
- Boundaries of acceptable modifications
- Validation protocol for each change type
- Performance monitoring plan

---

7. Cybersecurity for AI Models

7.1 AI-Specific Threats

Threat	Description	Risk	Mitigation
Adversarial attacks	Crafted input signals designed to fool the classifier	Medium	Input validation, anomaly detection, robustness testing
Data poisoning	Malicious data injected into training set	Low (locked model)	Training data provenance, quality controls
Model extraction	Reverse engineering the model through API queries	Low	API rate limiting, model obfuscation
Model inversion	Inferring training data from model outputs	Low	Differential privacy considerations
Evasion attacks	Inputs crafted to evade detection (e.g., VT that appears as NSR)	Medium	Robustness testing, confidence thresholds

7.2 Security Controls for AI Module

Control	Description
Model integrity	Checksum verification of model weights at load time
Input validation	Range checks, format validation, anomaly detection on input ECG data
Output validation	Confidence threshold (reject low-confidence classifications)
Access control	AI model API accessible only from authorized backend services
Audit logging	All inference requests and results logged
Model versioning	Cryptographic signing of model artifacts

Appendix A: GMLP (Good Machine Learning Practice) Compliance

GMLP Principle	Compliance
1. Multi-disciplinary expertise (clinical, technical, regulatory)	✅ Team includes cardiologists, ML engineers, regulatory specialists
2. Good software engineering and security practices	✅ IEC 62304, AAMI TIR57
3. Clinical study participants representative of intended population	⚠️ Ongoing improvement via PMCF
4. Training datasets independent of test datasets	✅ Strict split, no data leakage
5. Reference datasets based on best available methods	✅ Board-certified cardiologist annotations
6. Model designed to manage/minimize risks of overfitting, bias	✅ Dropout, regularization, cross-validation, bias analysis
7. Focus on the performance of the human-AI team	✅ Designed as decision support; usability tested
8. Testing demonstrate device performance under clinically relevant conditions	✅ Robustness testing + clinical validation
9. Users provided clear information about device performance and limitations	✅ IFU includes performance data and limitations
10. Deployed models monitored with retraining managed	✅ Post-market monitoring + change control

---

Control de Versiones:

Versión	Fecha	Autor	Descripción
1.0	2026-05-16	AI/ML Engineering	Creación inicial
